25901159 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Value Addition Course (VAC) - MG4VACCHE200 - BASIC ENVIRONMENTAL CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions

Each question carries **1** marks

1. The ozone layer is found in [K] / [CO1]

   a). Troposphere b). Stratosphere

   c). Mesosphere d). Thermosphere

2. Identify the technological factor that has contributed to increased global warming. [K] / [CO1]

   a). Energy-efficient appliances b). Industrialization using fossil fuels

   c). Solar power generation d). Wind energy utilization

3. Without the natural greenhouse effect, Earth's average temperature would be approximately: [U] / [CO1]

   a). 0°C b). -18°C

   c). 15°C d). 50°C

4. Excess anthropogenic greenhouse gases lead to: [U] / [CO1]

   a). Global cooling b). Global warming

c). Earthquakes d). Soil formation

5. Avoiding open burning of waste helps in [U] / [CO2]

a). Increasing air pollution b). Reducing air pollution

c). Increasing fog d). Increasing smog

6. The first stage of wastewater treatment is [U] / [CO2]

a). Secondary treatment b). Tertiary treatment

c). Primary treatment d). Disinfection

7. The main product formed when hydrogen is used as a fuel is [K] / [CO3]

a). Carbon dioxide b). Carbon monoxide

c). Water d). Methane

8. Which practice supports sustainable mineral utilization? [A] / [CO3]

a). Illegal mining b). Excess extraction

c). Metal recycling d). Open dumping

9. Ethanol from biomass is produced by [K] / [CO3]

a). Oxidation of cellulose b). Fermentation of sugars

c). Pyrolysis d). Hydrogenation

[1x9 = 9]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **4** marks

10. How does ozone behave as a greenhouse gas? [K] / [CO1]

11. What is sequestration of CO_2? [U] / [CO1]

12. What is ozone depletion? Mention any two causes of ozone layer depletion. [U] / [CO2]

13. What is acid rain? Mention any two causes of acid rain? [K] / [CO2]

14. List four benefits of biomass utilization in decarbonization. [K] / [CO3]

15. List four advantages of solar energy. [U] / [CO3]

16. Define mining.Discuss the purpose of mining. [U] / [CO3]

[4x5 = 20]

Part C

Short Essay Questions

Answer any **3** questions

Each question carries **7** marks

17. Explain the characteristics of thermosphere and exosphere. [K] / [CO1]

18. Explain in detail the process of removing CO_2 from the atmosphere using Direct Air Capture technology. [U] / [CO1]

19. Discuss in detail the major causes of water pollution. Classify the different categories of water pollution and explain each with suitable examples. [U] / [CO2]

20. Explain biomass energy and discuss various biofuels obtained from biomass. [K] / [CO3]

21. Discuss metallic and non-metallic mineral resources in detail, with examples and properties. [U] / [CO3]

[7x3 = 21]